

Felix Gröner

Human-LLM interaction researcher designing, developing, and evaluating LLM applications.

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Skills

Research: experiment design, human participants, data science, grant writing

Programming: Python, TypeScript, JavaScript, React, NextJS, SQL, Git

LLM systems: prompt engineering, LangChain, agent design, RAG

Education

Arizona State University, USA

Expected Fall 2026

Ph.D. Human Systems Engineering

Status: Ph.D. candidate, GPA 4.0

Committee: Erin K. Chiou (Advisor), D. Vaughn Becker, Nancy J. Cooke

Dissertation topic: Engineering and validating LLM agents as synthetic participants for Human Factors research

Ulm University, Germany

June 2024

M.Sc. Cognitive Systems

Thesis: The role of explainable artificial intelligence in trust and adherence to AI recommendations

Kiel University, Germany

December 2022

M.A. International Politics and International Law

Thesis: Necessity of adapting International Humanitarian Law to cyber warfare

Kiel University, Germany

December 2020

B.A. Computer Science and Political Science

Thesis: NATO's strategy to deter cyber attacks

Kiel University, Germany

December 2019

B.Sc. Business Informatics

Thesis: Client's influence on the choice of project management technique in software development

Professional Experience

Research

ADAPT Lab, Arizona State University

2025

Graduate Research Assistant

- Planned and conducted pilot studies and literature reviews
- Projects:

Semantic leakage in synthetic datasets

- Quantitative analysis identified biases in generated datasets affecting downstream evaluation validity.
- Applied mechanistic interpretability methods to identify the relevant attention heads. Ablating them did not reduce the issue, suggesting a deeper mechanism.

Human-LLM mutual competence modelling

- Designed and conducted a pilot study (n = 9) testing how well people and LLMs can assess their own and each other's likelihood of knowing the answer to the next question, given their performance history.
- Showed statistical reasoning patterns and high sensitivity to prompt framing.

Center for Accelerating Operational Efficiency

2023-2025

Graduate Research Assistant

- Wrote grant application and research proposals, authored papers
- Designed, developed, and deployed human-LLM interaction systems
- Projects:

Graphical user interfaces for directing expectations (\$132,000 awarded)

- Built a full-stack web application (NextJS, Postgres) that served as a custom experimentation platform for human-LLM interaction research, allowing researchers to control the UI/UX of and record behavioral metrics in experiments of human-LLM interaction.
- Composed a catalog of tasks that elicit misconceptions about LLM capabilities.

Reliance on LLM systems in intelligence analysis reporting tasks (\$210,000 awarded)

- Constructed an experimental setup in which participants (n = 304) solved an intelligence analysis task.
- Analyzed the data from different conditions that varied task difficulty, transparency, and stakes.

Human Factors research group, Ulm University

2022-2023

Undergraduate Research Assistant

- Designed and conducted research studies, project management
- Projects:

Explainable artificial intelligence (XAI) effects on adherence to AI recommendations

- Ran an experiment in which people (n = 404) saw either a global (linear surrogate) or local (LIME) explanation for an AI system's recommendation.
- XAI type did not have as large of an effect as the participants' prior self-perceived expertise.

Teaching

The Polytechnic School, Arizona State University

2026

Graduate Teaching Assistant

- Developed documentation for a full-stack (Python, Jinja2, MySQL) development capstone project

The Polytechnic School, Arizona State University

2023

Graduate Teaching Assistant

- Taught Python introductory classes
- Recorded and edited tutorial videos for Raspberry Pi

Publications

Peer-reviewed

Gröner, F., Verma, A., & Bronowitz, J. (2026). Automatically Generating Interactive Learning Experiences with an LLM Driven Agentic Pipeline. *Proceedings of the Learning Engineering Research Network Convening (LERN 2026): From Insights to Implementation, Learning Engineering in Action*. <https://doi.org/10.59668/2551.25387>

Gröner, F., & Chiou, E. K. (2024). Investigating the impact of user interface designs on expectations about large language models' capabilities. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. <https://doi.org/10.1177/10711813241260399>

Kraus, J. M., Merger, J., **Gröner, F.**, & Pätz, J. (2023). 'Sorry' says the robot: The tendency to anthropomorphize and technology affinity affect trust in repair strategies after error. *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (pp. 436-441)*. <https://doi.org/10.1145/3568294.3580122>

Preprints

Gröner, F., & Chiou, E.K. (2025). *Engineering approaches to improve human-LLM interaction: A scoping literature review*. <https://ssrn.com/abstract=6505859>

Gröner, F., & Becker, D.V. (2026). *Subjective and objective metrics to explain perceptions of original and AI-generated abstract artworks*. <https://ssrn.com/abstract=6482420>

Gröner, F., & Chiou, E.K. (under review). *A comparison of human and LLM-simulated participants in a writing style task.*

Gröner, F., Mancenido, M. V., Kim, N., Summers, E., & Chiou, E. K. (under review). *Increasing transparency of LLM systems does not always improve people's verification behavior and performance: Results from an empirical study of AI-assisted intelligence analysis reporting.*

Gröner, F., Mancenido, M. V., & Chiou, E. K. (in preparation). *Considering expertise as a factor in the design of synthetic test environments for human-AI research.*

Presentations

Gröner, F. & Becker, D.V. (2026). Two dimensions describe how we perceive original and AI generated abstract art. *2026 Cognitive Science Conclave*, April 25, Tucson, AZ.

Gröner, F. & Chiou, E. K. (2026). Caveats of replacing human participants with LLM simulations. *2026 IEEE Conference on Cognitive and Computational Aspects of Situation management (CogSIMA)*, March 9-12, Tempe, AZ.

Gröner, F. & Chiou, E. K. (2026). Engineering approaches to improve human-LLM interaction (Poster session). *AI Impact Pre-Summit @ ASU*, January 29, Mesa, AZ.

Chiou, E. K., Gröner, F., & Mancenido, M. V. (2025). Increasing transparency of LLM systems does not always improve people's verification behavior and performance: Results from an empirical study of an AI-assisted intelligence reporting task. Abstract submitted to the *69th HFES International Annual Meeting: ASPIRE*, October 13-17, Chicago, IL.

Gröner, F., Mancenido, M. V., & Chiou, E. K. (2024). Multi-source AI Scorecard Table (MAST) for evaluating generative AI in worker-automation team tasks. Project update presented at *Center for Accelerating Operational Efficiency (CAOE) Annual Meeting*, April 8-9, Washington, DC.